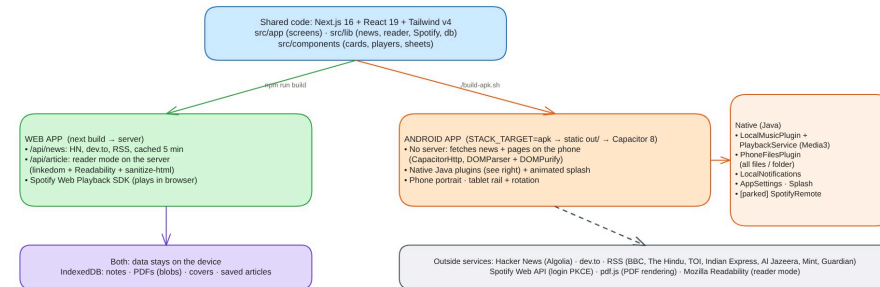




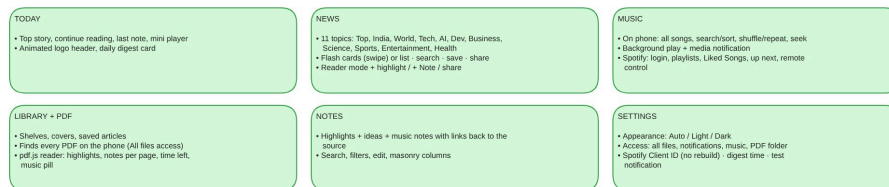
STACK — news · music · PDFs · notes in one app

Web app + Android app (Capacitor) from one Next.js codebase · status as of 28 Sep 2026 · v1.0

1 · How Stack is built



3 · What Stack does today (v1.0)

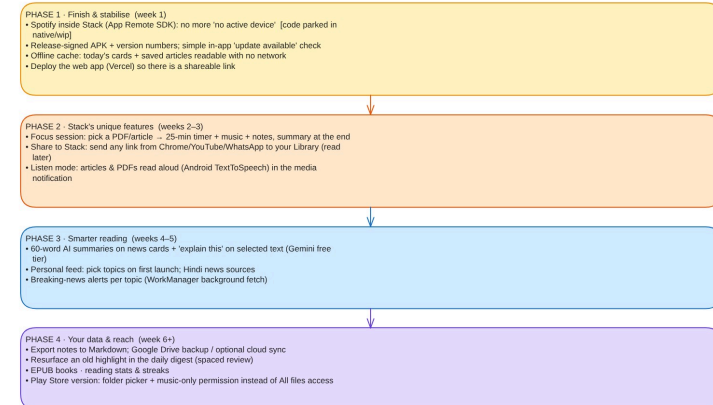


Tested on Android 16 emulators (Pixel phone + Pixel Tablet), light and dark. Web build + lint + type checks pass.

2 · Web app vs Android app: major differences

	WEB APP	ANDROID APP (APK)
How you get it	Open a URL (needs the Next.js server running / hosted)	Install Stack.apk (sideload, debug-signed), works without our server
News & reader	Fetches by our server (api), cached 5 min	Fetches directly by the phone (CapacitorHttp, no CORS)
Spotify	Can play inside the browser (Web Playback SDK, Premium)	Remote-controls the Spotify app, opens it if needed. In-app App Remote planned
Music on the device	Not available	MediaStore songs + Media3 background player, media notification, headset keys
PDFs	Pick files manually	Pick files, OR auto-find every PDF (All files access / one folder)
Notifications	None	Daily digest (local), permission flow in Settings → Access
Launch & branding	Favicon only	Animated logo splash, app icon, notification icon
Layouts	Phone column; wide layout on wide screens	Phone portrait; tablets rotate with the rail layout
Updates	Instant on redeploy	Rebuild + reinstall the APK (no auto-update yet)
Data	IndexedDB in that browser	IndexedDB in the app (not synced with the web)

4 · Plan: upcoming updates



5 · Sources for further research

Similar apps (competitors)

• Instants: 60 word news cards (Google Play) — <https://play.google.com/store/apps/details?id=com.nis.app>
• How Instants used AI (WAN-IFRA) — <https://wan-ifra.org/2020/03/how-instants-used-ai-to-become-one-of-indias-top-news-apps/>
• Readwise Reader: read-later, highlights, PDFs, TTS, AI — <https://readwise.io/read>
• Readwise Reader docs — <https://docs.readwise.io/reader>
• Best read-it-later apps 2025 (Mentle) — <https://www.getmentle.com/blog/best-read-it-later-apps-2025/>
• Omnivore alternatives 2026 (Readless) — <https://www.readless.app/blog/omnivore-alternatives-2026/>
• FocusTune: Pomodoro + kfi music (Google Play) — <https://play.google.com/store/apps/details?id=com.study.focustune.music>
• Focus Study: timer + notes (Google Play) — <https://play.google.com/store/apps/details?id=com.focustudy>
• ElevenReader: read-aloud reader — <https://elevenreader.io/>
• Best PDF readers for Android (Android Authority) — <https://www.androidauthority.com/best-pdf-readers-for-android-366394/>
• Best news apps 2026 (Zapier) — <https://zapier.com/blog/best-news-apps/>
• News aggregator apps compared (Daily Dev) — <https://daily.dev/blog/best-news-aggregator-apps-tested-compared/>

Build docs (for the next phases)

• Capacitor docs — <https://capacitorjs.com/docs/>
• Capacitor Local Notifications — <https://capacitorjs.com/docs/local-notifications>
• Spotify Web API — <https://developer.spotify.com/documentation/web-api>
• Spotify Android SDK (App Remote) — <https://developer.spotify.com/documentation/android>
• Android Media3 (background playback) — <https://developer.android.com/media/media3>
• Android: manage all files access — <https://developer.android.com/training/data-storage/manage-all-files>
• Android: receive shared content (Share to Stack) — <https://developer.android.com/training/sharing/receive>
• Android WorkManager (background news alerts) — <https://developer.android.com/topic/libraries/architecture/workmanager>
• Android splash screens — <https://developer.android.com/develop/ui/views/launch/splash-screen>
• Gemini API (AI summaries) — <https://ai.google.dev/gemini-api/docs>
• pdf.js — <https://mozilla.github.io/pdf.js/>
• Mozilla Readability — <https://github.com/mozilla/readability>
• Next.js static exports — <https://nextjs.org/docs/app/guides/static-exports>



WEB VS APP

Both versions are built from the **same Next.js code**. The web app runs with a server; the Android app is a self-contained APK (Capacitor) with native features. In short: **the web app is a reader you open in a browser; the APK adds everything that uses the phone itself**: music, files, notifications and background playback.

	WEB APP	ANDROID APP (APK)
How you get it	Open a URL; our Next.js server must be running or hosted	Install Stack.apk (sideloaded, debug-signed); needs no server of ours
News & reader	Our server fetches and cleans articles (/api), cached 5 min	The phone fetches pages itself (CapacitorHttp, no CORS limits)
Spotify	Can play inside the browser (Web Playback SDK, Premium)	Remote-controls the Spotify app and opens it when needed. In-app playback is planned
Music on device	Not available	Song library, Media3 background player, media notification, headset keys
PDFs	Pick files by hand	Pick files, or find every PDF on the phone (All files access / one folder)
Notifications	None	Daily digest; permissions managed in Settings → Access
Launch & brand	Favicon only	Animated logo splash, app icon, notification icon
Layouts	Phone column; navigation rail on wide screens	Phone portrait; tablets rotate with the rail layout
Updates	Instant on redeploy	Rebuild + reinstall (no auto-update yet)
Your data	IndexedDB in that browser	IndexedDB in the app (not synced with the web)

WHAT SHIPS TODAY

V1.0 · TESTED ON ANDROID 16 PHONE + TABLET EMULATORS, LIGHT & DARK

TODAY

- Top story, continue reading, last note
- Mini player, animated logo, daily-digest card

NEWS

- 11 topics incl. India, World, Tech, Sports
- Flash cards or list, search, save, share
- Reader mode + highlight / note

MUSIC

- Phone songs: shuffle, repeat, seek, background play
- Spotify: login, playlists, Liked Songs, remote control

LIBRARY + PDF

- Shelves, covers, saved articles
- Finds every PDF on the phone
- Highlights, notes per page, time left

NOTES

- Highlights, ideas, music notes linked to their source
- Search, filters, edit, masonry

SETTINGS

- Auto / Light / Dark theme
- Access: files, notifications, music
- Spotify Client ID, digest time



NEXT UPDATES

Four phases, ordered so that each one builds on the last. Phase 2 holds the features that no competitor combines.

PHASE 1 · FINISH & STABILISE

WEEK 1

- ☐ **Spotify inside Stack** with the App Remote SDK, so "no active device" never appears (code parked in `native/wip/`)
- ☐ **Release-signed APK** with version numbers and an in-app "update available" check
- ☐ **Offline cache**: today's cards and saved articles readable without network
- ☐ **Deploy the web app** (Vercel) for a shareable link

PHASE 2 · STACK'S UNIQUE FEATURES

WEEKS 2-3

- ☐ **Focus session**: pick a PDF or article → 25-min timer + music + notes → summary at the end
- ☐ **Share to Stack**: send links from Chrome, YouTube or WhatsApp straight into the Library
- ☐ **Listen mode**: articles and PDFs read aloud, with controls in the media notification

PHASE 3 · SMARTER READING

WEEKS 4-5

- ☐ **60-word AI summaries** on news cards and "explain this" on selected text (Gemini free tier)
- ☐ **Personal feed**: choose topics on first launch; Hindi news sources
- ☐ **Breaking-news alerts** per topic (background fetch with WorkManager)

PHASE 4 · YOUR DATA & REACH

WEEK 6+

- ☐ **Export notes** to Markdown; Google Drive backup or optional cloud sync
- ☐ **Resurface an old highlight** in the daily digest (spaced review)
- ☐ **EPUB books**, reading stats and streaks
- ☐ **Play Store build**: folder picker + music-only permission instead of All files access

RELATED FILES

Stack_Research_Sources.pdf: 25 sources (similar apps + build guides) · **ROADMAP.md**: this plan as a checklist · **Stack_App.excalidraw**: the editable diagram on page 1